Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/22/2007
PI Name:	Phillips, James O Ph.D.		
Project Title:	Gravitational Influence on Cerebellar Control of Gaze Movement and Adaptation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	98195	**Congressional District:**	7
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 5/2009) NOTE: Received NCE to 1/31/2008 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Fuchs, Albert (University of Washington) Robinson, Farrel (University of Washington)		
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Task Description:	Micro-gravity alters the normal relationships between coordinated multi-axial body movements and sensory feedback. As subjects undergo transitions between 1G to near 0G, they experience inappropriate eye and head movement and disorientation when shifting gaze. These changes pose a substantial challenge to mission objectives. One effort to address such challenges has been to develop context dependent preadaptation training protocols to train astronauts to quickly adapt based on inertial cues. Unfortunately, while this approach holds much promise, little is known about the reference frame for such adaptation or the underlying physiological mechanisms responsible for these adaptive changes. This black box approach limits progress toward the development of appropriate countermeasures. Our approach is to develop a laboratory analog of the altered gravitational/inertial environments, and then use identical psychophysical tests to evaluate behavioral performance of human and animal subjects in these environments to define both the neural mechanism and the crucial factors determining adaptation. Our research will consist of 3 components. First, we will describe the characteristics of context dependent human adaptation and its transfer to other forms of adaptation, seeking to localize behaviorally the site of adaptation in the CNS. Second, we will perform identical experiments on rhesus monkeys while recording the neural activity of cells within the caudal fastigial nucleus, which is thought to be important for saccadic adaptation. Third, we will simulate changes in the vestibular environment by changing the activity of the vestibular end organ in monkeys. We will study context dependent adaptation and the neural discharge associated with adaptation in this altered vestibular environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has a significant impact on life on earth because it contributes to our knowledge of how the brain controls movement and adapts those movements to challenges faced by novel environments, injury and disease. Eye movements are critically important to everyday human activity because the eye must continually scan the visual world for us to see the world clearly. These eye movements must be fast and accurate, and the orientation of the eye must be maintained in the face of self motion and motion of objects in the visual environment. Similarly, the vestibular sensory system performs a critical function in allowing us to remain stable and oriented in the world. These systems are constantly adapted to allow them to perform their critical roles, and this research addresses the nature of the neural mechanisms that underlie that adaptation. In addition, adaptation procedures developed in this grant may be useful in promoting recovery of behavioral and sensory function following injury. Also, technology derived from this research, including new clinical eye measurement technology and more recently approaches to develop and evaluate a vestibular neural prosthesis will benefit patients with oculomotor and vestibular disorders.
	We have made continued progress on each aspect of our proposed research project in the third year of approval and we anticipate completing the entire project by the end of January 2008. Our human data collection is completed, but our analysis of the human data is continuing. Our data collection in rhesus monkeys is continuing, and we anticipate that this will be completed in the fall of 2007. Our progress has resulted in the submission of several manuscripts for publication in year 3. One manuscript was published as a paper in Vision Research, several are undergoing revision and one is currently submitted. Each group working on this project has contributed significant findings. In addition, in related activity, we have pushed forward with a new technology developed under this research project for eye and head movement recording. We have received a fully funded R21 research grant from the NIDCD to continue development of this technology. Furthermore, we used preliminary research results from this project on the effects of modification of canal afferent input to design and submit a response to an NIH Neural Prosthesis RFA. The contract, which was subsequently awarded to our group in FY2006 will utilize the techniques developed in our project to evaluate a new implant technology to replace or modify input from the semicircular canals and otolith organs. One exciting application of this technology would be to modify canal input to facilitate adaptation to microgravity. If the vestibular input can be adjusted to provide coherent vestibular inputs that mimic those of 1G during spaceflight, adaptation to this novel environment might be enhanced (i.e., made more rapid and complete). Finally, we continue our collaboration with the group of Dr. Millard Reschke at Johnson Space Center, Houston, on the interpretation of human saccadic eye movement data collected before, during, and immediately following spaceflight. Our progress for the third year is documented below. Human Experiment - Year 3 update. (Fuchs, Ling, Hopp, and Phillips) – Our results confirm that eye and head movements are adapted together during head unrestrained gaze adaptation in humans. We have also shown that adaptation of human saccadic eye movements in the head restrained laboratory condition transfers to the head unrestrained condition, and adaptation of head unrestrained gaze movements transfers to the head restrained condition. Furthermore, the transfer shows the same "adaptation field" in both the head unrestrained and head restrained conditions. Also, the adaptation field is in gaze sensory coordinates, and moves with changes in the orientation, tilt, of the head. This result has important operational significance because it suggests that a human preflight saccade adaptation countermeasure will transfer to the context in which natural head unrestrained gaze movements are being performed. That transfer will be in the sensory coordinates of the initial gaze (or retinal) error. Our data has been analyzed and a manuscript has been prepared from this research. Human Experiment - Year 3 update. (Fuchs, Ling and Phillips) Our results show that changes in orientation including roll tilt and inversion of subjects have significant effects on the kinematics of saccadic and smooth eye movements in the horizontal or vertical plane. In particular, inversion of human subjects produces consistent decreases in saccade precision, and variable changes in accuracy. Smooth pursuit shows an increase in saccadic intrusions, and decreases in gain in many subjects during inverted viewing of sinusoidally moving targets. No consistent changes were observed in horizontal optokinetic nystagmus, although vertical OKN was affected. These results are interesting in that they imply that human eye movements in both the horizontal and vertical plane are affected by changes in whole body orientation. This data is currently undergoing analysis and a manuscript is in preparation. Human Experiment - Year 3 Update. (Fuchs and Hopp) – This work has resulted in a manuscript, which is currently undergoing review . The manuscript, titled "Identifying sites of saccade amplitude plasticity in humans: transfer of adaptation between different types of saccade" reports that amplitude reductions in memory-guided, delayed, targeting and express saccades transfer in various amounts to other saccade types. For example, when memory-guided saccades were adapted, the amplitude of express, targeting and delayed saccades were partially adapted. In contrast, when express, targeting, or delayed saccades were adapted, there was complete transfer of adaptation to memory-guided saccades. These results, combined with other data, suggest that there are at least two sites of adaptation within the

Task Progress:

saccadic system. One is used commonly in the generation of express, targeting, delayed and memory-guided saccades, whereas the other is specific for the generation of memory-guided saccades. The relevance of this work is that it informs operational implementation of a preflight adaptation countermeasure.

Human Experiment - Year 3 update. (Fuchs and Hopp) This work has resulted in a published paper. The paper, titled "Amplitude adaptation occurs where a saccade is represented as a vector and not as its components" reports that human saccade adaptation occurs where the saccade command is represented as a vector, and not its horizontal and vertical components. In one experiment, we behaviorally reduced the amplitude of clockwise oblique saccades and examined the transfer to saccades made to other target amplitudes and directions. In a second, we adapted rightward saccades of the same size as the rightward component of the clockwise oblique saccades and examined the effect on oblique saccades. The relevance of this work is that it defines the location of the site of adaptation. This data is consistent with a site of adaptation that is upstream of the brainstem burst generator.

Animal Experiment - Year 3 update. (Phillips, Fuchs, Ling, Newlands). This work has resulted in a manuscript that is currently in a second revision. In year three, we recorded from an additional animal in which modulation of canal afferent input was surgically altered. This work was featured in an American Neurotology Society panel symposium on Surgical Partitioning of the Labyrinth. We reported that surgical modification of canal afferent input modifies the reflex behavior and discharge of brainstem neurons during head unrestrained gaze shifts in rhesus monkeys. We observed that animals made accurate gaze shifts following canal plugging, but the relationship of saccadic eye movement end and gaze movement end, and the amplitude and duration of the eye and head movement components was dramatically altered. This data suggests that vestibular signals are important for the normal control of gaze movement, but that new strategies emerge, involving adaptation of saccade amplitude, to compensate for the loss of vestibular input and maintain accurate gaze movements after canal plugging. Cells in the brainstem burst generator and vestibular nuclei also alter their discharge properties. An efference copy of the head movement may emerge in the discharge of brainstem burst neurons, but not in vestibular nucleus neurons, to produce the observed eye counterrotations during gaze shifts These data are critical for assessing the importance of vestibular signals during gaze shifts with normal and abnormal vestibular contributions. In addition, they suggest that the signature of brainstem saccadic eye movement related neural activity is dramatically altered by loss of vestibular input, leading to a the emergence of new mechanisms for accurate gaze control.

Animal Experiment. Year 3 update (Fuchs, Ling, and Phillips) We have now recorded from a subset of neurons in the caudal fastigial nucleus (CFN) of 3 rhesus monkeys that display discharge characteristics consistent with a head movement signal during active head unrestrained gaze shifts. This discharge is similar to the head related signal of prolonged burst neurons recorded in the pontine brainstem during head unrestrained gaze shifts, in that it dramatically outlasts the saccade related discharge of saccade neurons in the CFN. Such a neuron could carry the vestibular signals requisite for the control of coordinated gaze movement during head unrestrained gaze shifts. Likewise, these cells are possible targets for the vestibular inputs that may be required to establish the context for context dependent adaptation. We continue recording from this population during both head unrestrained gaze shifts and tilt context dependent adaptation.

Animal Experiment. New for Year 3. (Fuchs, Ling, and Phillips) We have now recorded from neurons in the cerebellar nuclei subjacent to the dentate nucleus, and lateral to the CFN, that show a discharge for saccadic eye movement and head unrestrained gaze movement. These cells, which have been recorded in a single animal to date, show a burst discharge in advance of saccades. In addition, some of these neurons, located more dorsally, show a modulation with head velocity during head unrestrained gaze shifts. Injection of mucimol in this region of the cerebellum produces both a change in the gain and accuracy of saccades, tilt of the trunk and head, and motor incoordination of the upper limbs. Discovery of this region was entirely serendipitous, as we were exploring the lateral extent of oculomotor related activity in the cerebellar nuclei in association with CFN recording experiments. In the final year of this research activity, we plan to complete a study of this area in another monkey. In addition, we have submitted an abstract describing these results to the Society for Neuroscience annual meeting. This result is important, in that it identifies a potential site for the neural control of eye, head, and hand coordination. If this region is involved with coordinated adaptive control of these motor systems, it would represent an important target for pre-flight counter measures to facilitate adaptation of critical coordinated task performance during spaceflight (such as coordination of eye, head and hand movements during launch and reentry maneuvers).

Animal Experiment. Year 3 Update (Noto, Robinson and Phillips) This work has resulted in a manuscript that is undergoing revision. In rhesus monkeys, saccades made during tilt orientations in pitch and roll are dissymmetric, producing dissociated vertical deviation of the eye and, in some monkeys, changes in saccade velocity and amplitude. This data suggests that vertical skew deviation does accompany roll tilt of approximately 45 degrees in monkeys during viewing of near point visual stimuli. This could pose a significant challenge to mission objectives if similar dissociations are present in humans under similar conditions.

Animal experiment. Year 3 update. (Phillips, Noto and Robinson) We previously determined, in rhesus monkeys, that the adaptation field during tilt is in craniotopic coordinates and is largely unaffected by tilt orientation. In addition, we found a similar phenomenon in human subjects subjected to the same testing paradigm. We are still analyzing the human data, and collecting additional data in rhesus monkeys. This result confirms an important limitation to preflight adaptation paradigms as a countermeasure. Since the adaptation is restricted to specific vectors with respect to the head, the range of vector directions must be extended during preflight training to allow for adequate representation of visual space during natural movements.

Animal experiment - Year 3 update. (Phillips, Noto and Robinson). Rhesus monkeys display transfer of adaptation across tilt adaptations, and initial tilt orientation does affect to degree of transfer between orientations. The rate constants for adaptation were faster in the head upright condition. We have completed data collection in studies of the transfer of adaptation in human subjects. This adaptation does transfer in humans, suggesting that adaptation-training countermeasures will transfer to a range of tilt orientations or inertial contexts. Data analysis is continuing on this study.

Animal Experiment - Year 3 update (Phillips, Noto, Ibarreta, Robinson, and Andrews). Context dependent adaptation has been performed in rhesus monkeys during pitch and roll tilt. This data was presented during year three at the Washington Research Symposium in short paper titled " Using eye movement research to prevent space sickness". We reported that roll cues are more salient than pitch cues for context dependent adaptation. In addition, difference exist for forward and backward pitch adaptation which are context independent. Since roll tilt produces differential ocular torsion in the two eyes and pitch cues do not, it is possible that oculomotor torsion provides the context for context dependent

	adaptation and not the vestibular input per se. This result may have profound implications for the application of context dependent saccade adaptation paradigms as a countermeasure, since the implication is that the primary context cue is oculomotor, not vestibular. Animal Experiment -Year 3 update. (Phillips, Noto, Andrews, and Robinson). We continue recording vestibular and saccadic responses in the CFN of monkeys that are subjected to roll tilt. Our analysis is currently ongoing. Related activity. (Kaneko, Phillips) As an extension of our year two activity in which we developed with a local firm a printed circuit device that could serve as a cheap alternative to the existing eye movement transducers, in year 3 we received NIDCD funding in the form of an R21 titled " Vestibular and optokinetic testing for research and clinic" to continue this work. Dr. Kaneko is the principal investigator and Dr. Phillips is the Co-P.I. on this grant. Dr. Phillips is currently working with NeuroConnex Corp. on an SBIR application for development of a clinical grade version of this device. This funding and continued development of this technology is a direct outgrowth of our experimental research funded through this project. Related activity. (Reschke, Phillips) Our ground based research compliments flight research begun many years ago at Johnson Space Center by Reschke, Harm, Paloski, Krnavek, Somers, and Ford. Dr. Reschke recorded pre-flight, flight and post-flight saccadic eye movement data in astronauts during several space missions. We are analyzing this very important flight data to better understand the nature of eye movement changes related to adaptation to novel inertial environments. This data analysis is currently ongoing. Related Activity: (Phillips, Fuchs, Ling). An important outgrowth of our research activities and a new research direction for our laboratory was the submission and subsequent funding of an NIH / NIDCD contract proposal titled "Neurophysiological studies of electrical stimulation for the vestibular nerve" to develop a vestibular prosthesis based on existing cochlear implant technology. Dr. Phillips is P.I. on the contract, which will utilize the approaches described and utilized in this project, including canal plugging and gentamicin injection, and neural and behavioral recording of adaptive recovery of vestibular function during natural movements in rhesus monkeys. This proposal would not have been possible without the knowledge and preliminary results acquired through this NASA funded research. The development of a neural prosthesis has potentially significant implications for the neurovestibular countermeasures program at NASA, because this technology could ultimately be used to tune vestibular responses to the unique inertial and gravitation environments encountered in 0G spaceflight. In addition, the contract will support studies that will further our understanding of the neural mechanisms underlying vestibular adaptation, and in that regard, are a continuation of this NASA funded activity.
Bibliography Type:	Description: (Last Updated: 05/13/2009)
Articles in Peer-reviewed Journals	Hopp JJ, Fuchs AF. "Amplitude adaptation occurs where a saccade is represented as a vector and not as its components." Vision Res. 2006 Oct;46(19):3121-8. Epub 2006 May 15, PMID: 16698057 , Oct-2006